

AI & Machine Learning – Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

1. Objective

Task 3 advances beyond Task 2 by addressing model reliability. The focus shifts from simply training multiple models to detecting overfitting, validating performance using cross-validation, and systematically tuning hyperparameters using GridSearchCV — mirroring professional ML workflows.

2. Dataset

The **California Housing Dataset** (20,640 samples, 8 features) was used — identical to Task 2 for fair comparison. Target: Median House Value (HousePrice). Features: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude.

3. Step-by-Step Methodology

Steps 1–4: Setup (Same as Task 2)

Libraries imported, dataset loaded, features scaled with StandardScaler, and data split 80/20 (train/test, random_state=42) for reproducibility.

Step 5: Overfitting Detection

An unconstrained Decision Tree (no max_depth) was trained. Train RMSE = 0.0 vs Test RMSE = 0.4054 — a gap of 0.4054 confirms the model memorised training data and failed to generalise. This is a classic overfitting signature.

Step 6: Cross-Validation (5-Fold)

cross_val_score with cv=5 and scoring='neg_root_mean_squared_error' was applied to the full tree across the entire dataset. Result: CV RMSE = 0.3998 ± 0.0027 . The low standard deviation confirms the estimate is stable and not split-dependent. Cross-validation is preferred over a single split because it uses all data for evaluation.

Step 7: Hyperparameter Tuning — GridSearchCV

GridSearchCV tested all combinations of max_depth $\in \{3, 5, 7, 10\}$ and min_samples_split $\in \{2, 5, 10\}$ (12 configs $\times$ 5 folds = 60 model fits). Best configuration: max_depth=10, min_samples_split=10. Best CV RMSE: 0.3415.

Step 8: Tuned Model Evaluation

The best estimator from GridSearchCV was evaluated on the held-out test set. Train RMSE = 0.2466, Test RMSE = 0.3376, $R^2 = 0.9435$. The train-test gap shrank from 0.4054 to 0.091 — overfitting is controlled.

Step 9: Model Comparison Summary

All three models were compared side-by-side on identical test data using RMSE and R^2 . See results table in Section 4.

Step 10: Final Model Justification

Model selection was made using measurable criteria — see Section 5.

4. Results

Model	Train RMSE	Test RMSE	R ² Score	Notes
Linear Regression	0.7256	0.7256	0.739	Baseline — linear only
Ridge Regression	0.7256	0.7256	0.739	Same as LR; no multicollinearity
Full Decision Tree	0.0	0.4054	—	Severe overfitting
Tuned Decision Tree	0.2466	0.3376	0.9435	Best — tuned & generalised

5. Final Model Selection Justification

Selected Model: Tuned Decision Tree Regressor (max_depth=10, min_samples_split=10)

Why this model was selected:

Highest R² (0.9435) and lowest Test RMSE (0.3376) across all models. It explains 94.3% of variance in house prices — substantially better than linear models (~73.9%).

How overfitting was reduced:

The full tree had a train-test RMSE gap of 0.4054. After tuning, this gap fell to 0.091 — a reduction of over 90%. max_depth limits tree growth; min_samples_split prevents splits on tiny leaf groups.

Why cross-validation results are trusted:

5-fold CV RMSE = 0.3998 ± 0.0027 . The standard deviation is very low, confirming that performance is consistent across different data subsets — not a lucky outcome from one particular split.

Trade-offs considered:

Decision Trees are less interpretable than linear models. However, the significant performance improvement justifies the complexity. Ridge and Linear Regression produced identical results because the data lacks strong multicollinearity, so L2 regularization added no benefit in this case.

6. Key Learnings

- Overfitting is detected by comparing train and test RMSE — a large gap is a red flag.
- Cross-validation gives a more reliable and stable performance estimate than a single split.
- GridSearchCV automates hyperparameter search and uses CV internally for fair comparison.
- max_depth and min_samples_split are the primary controls for Decision Tree overfitting.
- Model selection must be data-driven using metrics, not intuition.

Appendix: Performance Charts

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